A Novel UWB-Based Full-Range and Multi-Angle Following Robot

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***Abstract*—In recent years, research on elderly assistance and personal bodyguard robots has gained significant attention, with the global population aging and social security concerns intensifying. This highlights the vital role of person-following robots in these contexts. However, current robots typically operate within a limited range of 1.0m to 2.0m, which is insufficient for elderly care scenarios that require closer proximity for safety and responsiveness. Additionally, some older individuals prefer long-distance following for privacy reasons. Moreover, dynamic environments, such as changes in spatial crowding, demand robots capable of multi-angle following for enhanced adaptability. These challenges raise a key scientific problem: can we design a full-range, multi-angle system that meets the safety, rapid response, and privacy requirements of elderly care robots while adapting to variable environments? To this end, we propose a novel ultra-wideband (UWB)-based full-range and multi-angle following system for elderly care. Our system incorporates algorithms for both close-range and long-distance following, along with a mechanism that facilitates a seamless transition between the two modes. For close-range following, we develop a multi-sensor fusion algorithm to achieve both precise distance control and multi-angle following. For long-distance following, we implement automatic range adjustment, fall detection recall, and active recall mechanisms to ensure effective monitoring and responsiveness to unexpected events. Besides UWB, the full-range following system also incorporates 2D LiDAR, wireless communication, and inertial measurement unit (IMU) to enhance system reliability. Extensive experiments validate the effectiveness of our system across various distances and scenarios, demonstrating its potential to enhance both safety and privacy in elderly care applications.**

***Index Terms*—person-following robot, close-range following, long-range following, elderly care.**

I. Introduction

In recent years, the quality of life and safety of the elderly have become major societal concerns. Recent statistics [1] indicate that the proportion of people aged 65 and older in the global population continues to increase, a trend that is expected to intensify in the coming decades. Thus, the care of older adults has become a global social challenge, with a growing and urgent need for monitoring systems that can offer timely assistance and support to older adults [2], [3]. With this trend, robotic technologies, particularly person-following robots, have emerged as a promising solution for continuous monitoring and assistance in elderly care.

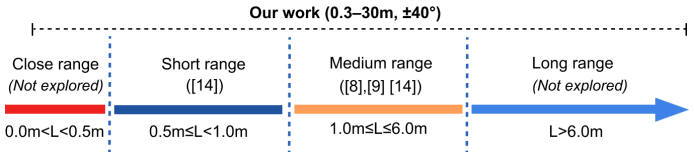


Fig. 1. Current research predominantly covers short to medium ranges (0.5m–6.0m), with little to no exploration of close-range (<0.5m) or long-range (>6.0m) following. Our system extends this gap, offering full-range and multi-angle following capabilities (0.3m–30m, ±40°).

Considering the varying requirements of different application scenarios, we categorize human–robot following distances into four ranges: close range (0.0–0.5 m), short range (0.5–1.0 m), medium range (1.0–6.0 m), and long range (over 6.0 m), as illustrated in Fig. 1. This classification corresponds to typical scenarios, including elderly assistance (0.0–0.5 m) [4]–[6], shopping assistance (0.5–1.0 m), highlight-detection sports robots (1.0–6.0 m) [7], and privacy-preserving monitoring (over 6.0 m). Existing research on following robots primarily focuses on maintaining distances of 1.0–2.0 m [8], [9] and rarely supports multi-angle following. Consequently, close-range assistance (0.0m-0.5m), which is critical for tasks such as helping an elderly person walk or providing immediate support, remains largely unexplored in the context of person-following robots. Similarly, long-range following (over 6.0 m), which is important for privacy-preserving and wide-area monitoring, remains largely unexplored. Furthermore, all existing approaches lack the capability to perform both full-range and multi-angle following concurrently.

Based on such motivations, this study focuses on designing a full-range, multi-angle following robot to meet the requirements of safety, rapid response, and privacy in elderly care. To the best of our knowledge, this is the first system that simultaneously supports 0.3–0.7 m tight following, ±40° multi-angle following, and 10–30 m long-distance recall in a single unified UWB-centric architecture for elderly care robots. Our approach adopts the full spectrum of following distances within a single cohesive system, addressing the research gap between close and long-range following. It also ensures compatibility with existing short-range and medium-range methodologies, as illustrated in Fig. 1. For close-range

following, we implement a multi-sensor fusion approach based on UWB and 2D LiDAR. The UWB-LiDAR solution provides precise distance control, continuous target tracking, and robust adaptability across diverse environments. It supports multi-angle following, enabling target tracking at preset angles between -40° and 40° while maintaining a following distance within 0.3 to 0.7 meters. This capability enhances safety and flexibility, making it well-suited for dynamic and complex care scenarios. In long-distance following, our system introduces innovative features such as automatic range adjustment, fall detection triggers, and active recall mechanisms. Using the long-range positioning capabilities of the UWB and the synergy of wireless communication devices, the system can acquire real-time target location information and ensure stable signal transmission. These features allow the robot to maintain effective monitoring at greater distances while remaining responsive to unexpected events or changes in the elderly person's needs. Meanwhile, we implement a novel mode-switching mechanism between close-range and long-distance following modes, addressing the diverse needs in elderly care scenarios.

The contributions of this work can be summarized as follows:

- We present a novel full-range and multi-angle following approach, fully integrated into a single cohesive system, that effectively addresses the multifaceted requirements of elderly care, from close companionship to privacy-preserving distant monitoring.
- For close-range following, we propose a UWB-guided LiDAR clustering algorithm that enables precise control and multi-angle tracking, ensuring flexibility and safety in elderly care.
- We propose a scenario-based recall module for long-distance following that features a target-triggered recall mechanism and automatic range adjustment, while balancing privacy protection and monitoring requirements in elderly care.
- The empirical results validate the effectiveness of our system, improving the adaptability and reliability of elderly care robots in various scenarios.

II. Related Work

During the past decade, person-following robots have seen increasing applications in elderly care [4]–[6], [10], providing valuable assistance and interaction to elderly and people with special needs [11]. Traditional research on person-following robots has focused on distances between 0.5m-6.0m [8], [9], [12], [13]. However, close-range assistance (0.0m-0.5m), which is critical for tasks such as helping an elderly person walk or providing immediate support, remains largely unexplored in the context of person-following robots. Several studies focus on close-range following robots. For example, Ye et al. [14] proposed a visible-joint-based person following robot with a following range 0.5m to 6.0m, which effectively addresses issues caused by partial occlusion. Nevertheless, when the following distance is reduced to below 0.5 meters, only the knees are visible, and AlphaPose's low robustness in detecting moving knees results in unreliable localization, potentially leading to safety concerns.

Privacy protection for the elderly care robot has been extensively studied [15]–[17]. These studies highlight the importance of preserving privacy in assistance robots and emphasize the need for privacy safeguards. However, long-distance following assistance robots, which could address some privacy concerns and wide-area monitoring, remain understudied. Moreover, current research on person-following robots focuses primarily on side-by-side or rear-following modes [11], [18]–[20]. It lacks support for multi-angle following, which limits adaptability to various care scenarios.

Several studies have explored combining ultra-wideband (UWB) ranging with LiDAR sensing for person-following applications. Zhai et al. [21] present a person-following robot that integrates UWB measurements with a single-plane 2D LiDAR, achieving improved SLAM map quality and trajectory optimization. More recently, [22] presented a UWB-based human-following system that uses LiDAR-derived terrain features to avoid obstacles and crevasses, enabling exploration in extreme environments. In all these efforts, UWB provides absolute distance measurements while LiDAR captures relative obstacle geometry. In contrast, we introduce a novel algorithm that uses UWB-guided LiDAR point-cloud extraction for person following, enabling precise distance control and improved safety throughout the tracking process.

The reviewed literature reveals no research on full-range, multi-angle following systems tailored for elderly care. To fill this gap, we present a novel system that integrates close-range, long-distance, and multi-angle following capabilities within a single framework.

III. System Design

A. System Architecture

Our full-range, multi-angle following platform integrates UWB (UWB-X2-AOA-N base / UWB-X2-AOA-WB tag), two 2D LiDARs (RPLiDAR A3M1, YDLiDAR G2), an ultrasonic sensor, IMU, Bluetooth/wireless links, a mobile device, a high-performance PC (ROS Noetic; interchangeable with Jetson), and an AGV, as shown in Fig. 2. The upper LiDAR (mounted at $\sim$1.2 m) collaborates with the UWB base to extract the target and regulate distance in real time, while the lower LiDAR handles obstacle sensing and avoidance. This complementary layout mitigates leg occlusion and point-cloud fragmentation, reducing association dropouts and improving robustness. The hardware is modular: compute can be replaced by an embedded board, and LiDARs by lower-cost units provided their range and scan rate meet our minimum specifications. Additionally, the software architecture is illustrated in Fig. 3.

B. Close-Range Following

1) The Novel UWB-LiDAR Fusion for Robust Target Tracking: Unlike prior UWB–LiDAR fusion methods—either UWB-assisted LiDAR SLAM [23], [24] or UWB-augmented

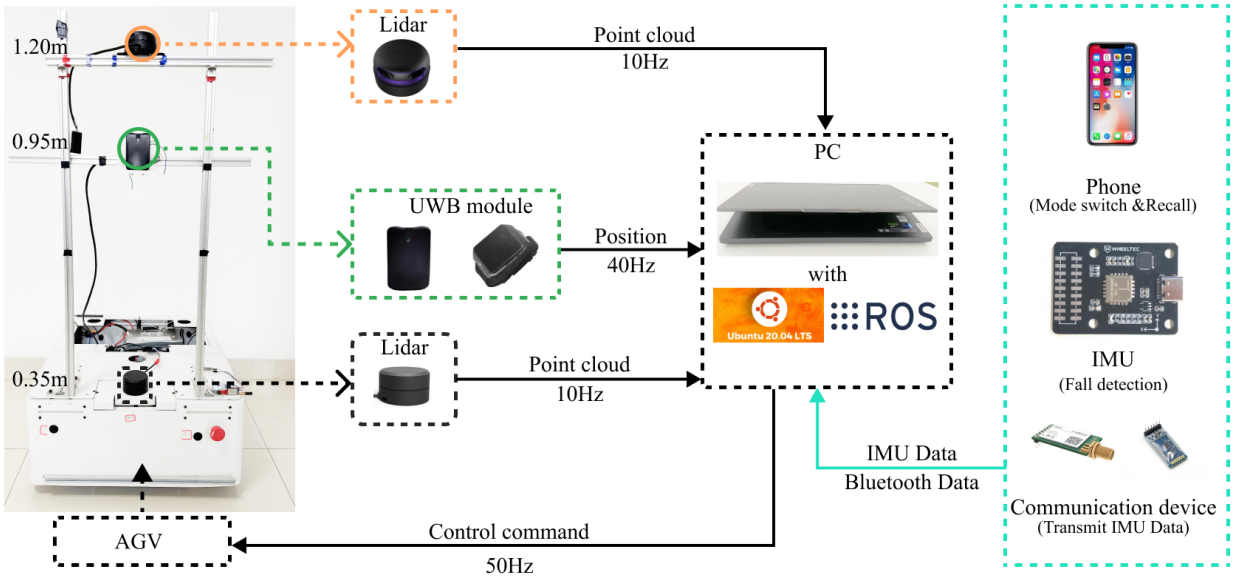


Fig. 2. Architecture of the full-range and multi-angle following system. This figure illustrates the system architecture, integrating close-range and long-range following capabilities. It integrates key hardware components including a UWB positioning module, two 2D LiDAR sensors (for distance control and obstacle avoidance), a Bluetooth module, a wireless serial communication module, an AGV, a PC, and a mobile phone for mode switching.

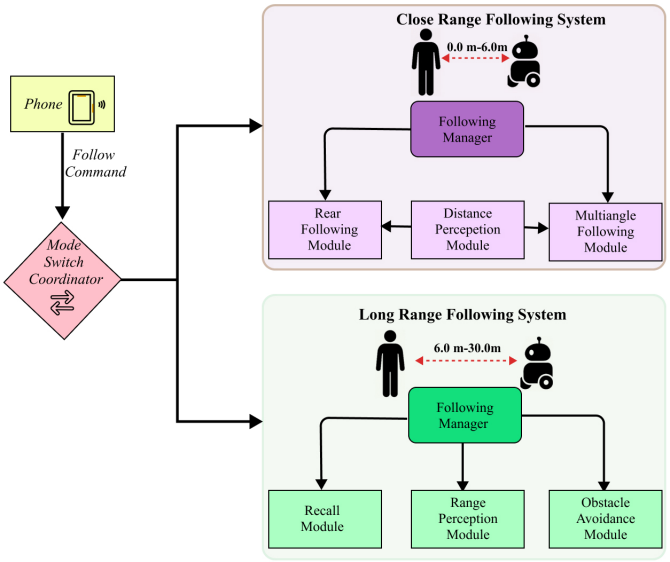


Fig. 3. Architecture of the Dual-Mode Following System. This figure illustrates the software architecture of the dual-mode following system. It comprises a mode-switch module that monitors Bluetooth signals to determine the follow flag, dynamically activating either the close-range or long-range following system.

person-following for localization/mapping and obstacle avoidance [21], [25], We introduce a novel UWB-guided LiDAR clustering algorithm, a key insight for robust target tracking in elderly care scenarios. Our method's uniqueness lies in its synergistic fusion of UWB localization and adaptive LiDAR point-cloud processing, overcoming inherent limitations in single-sensor approaches. Our innovative use of UWB data to guide LiDAR processing represents a significant advance in multi-sensor fusion. We leverage UWB-provided distance and angle to estimate the target's position, constraining LiDAR data processing to a specific angular range ($[\frac{-\pi}{2}, \frac{\pi}{2}]$). The algorithm identifies the target by comparing cluster centroids to the UWB-estimated position, effectively filtering non-target objects. The controller maintains a stable following distance by synchronizing the robot's speed with the target's walking speed, using 2D LiDAR data as input.

2) Flexible Multi-Angle Robot Following Algorithm: We introduce an innovative multi-angle following algorithm that significantly enhances the flexibility and adaptability of human-following robots in elderly care scenarios. Formally, we formalize multi-angle following in a human-centric polar frame: the robot regulates its relative bearing and radius to prescribed values within the human's rear sector—that is, at bearings $\theta \in [-40°, +40°]$ measured about the human, with $\theta = 0°$ denoting directly behind the human. Unlike previous works [12], [20], which are limited to direct rear following, our approach enables following at user-defined angles, representing a substantial advancement in assistive robots. Our algorithm dynamically adjusts the robot's trajectory based on real-time UWB position data and a preset following angle, while utilizing LiDAR data for distance control. The system's ability to maintain stable tracking across various preset angles, as demonstrated in Algorithm 1, offers unprecedented flexibility in human-robot interaction scenarios.

C. Long-Distance Following

We design a novel long-distance following method to enhance the privacy protection of the target and meet diverse needs. This section consists of three main components: adaptive range adjustment, dual-mode recall mechanism, mode-switching mechanism.

Algorithm 1 Flexible Multi-Angle Robot Following

Input: UWB data $(d_{\mathrm{uwb}},\ \theta_{\mathrm{uwb}})$;
LiDAR point cloud P;
Desired following distance d_{dsr};
PID gains $(K_{\mathrm{p}},\ K_{\mathrm{i}},\ K_{\mathrm{d}})$
Output: Control command C_{cmd}

while the ROS node is running **do**
  $(x_t, y_t) \leftarrow \mathrm{TagPose}\big(z_t^{\mathrm{uwb}}\big)$
  $P_t^{\mathrm{fov}} \leftarrow \{\, p \in P_t \mid \angle(p) \in \mathcal{F} \,\}$ *(frontal FOV selection)*
  $\varepsilon_t \leftarrow \mathrm{Clusterparam}\big(P_t^{\mathrm{fov}}\big)$
  $\mathcal{C}_t \leftarrow \mathrm{EuclidCluster}\big(P_t^{\mathrm{fov}}, \varepsilon_t\big)$
  $P_t^{\star} \leftarrow \arg\min_{C \in \mathcal{C}_t} \big\|\mathrm{Centroid}(C) - (x_t, y_t)\big\|$
  $d_t \leftarrow \big\|\mathrm{Centroid}\big(P_t^{\star}\big)\big\|$ *(current following distance)*
  $v_t \leftarrow \mathrm{PID}_r\big(d_t - d_{\mathrm{dsr}}\big)$ *(linear velocity from radial error)*
end while

1) Adaptive Long-Distance Accompaniment System for Privacy: Our long-distance accompaniment system introduces an innovative standby mode for the Automated Guided Vehicle (AGV), activated via mobile device, to balance privacy preservation with safety monitoring. Utilizing Ultra-Wideband (UWB) technology, the AGV continuously tracks the target's real-time location, measuring both distance and angle. A predefined safety-following range is established, within which the AGV maintains vigilant monitoring. When the target exceeds this range, the AGV initiates autonomous position adjustment, re-establishing the target within the specified boundaries before resuming its distant standby state. This dynamic approach significantly enhances system reliability and practicality, optimizing the effectiveness of long-distance accompaniment in elderly care scenarios.

2) Dual-Mode Recall Mechanism for Remote Elderly Monitoring: Our dual-mode recall mechanism is executed by a ROS-based state machine in a closed loop: (i) trigger (active summon or fall event), (ii) UWB-based target localization, (iii) obstacle-aware path planning and control, and (iv) adaptive-speed navigation to rendezvous. In active mode, an Android app issues a Bluetooth recall and the AGV navigates to the target using UWB localization with obstacle avoidance. In fall-triggered mode, a threshold-based IMU monitors body orientation; this detector is modular and can be replaced by standard fall-detection methods. When critical thresholds are exceeded, the system switches to an emergency approach mode. Safety is enhanced by a front-mounted LiDAR with an adaptive, multi-threshold scheme that cascades safety margins and modulates speed by obstacle proximity, improving robustness and efficiency for long-range accompaniment.

3) Multiple Following Mode Switching: To enhance the flexibility of our following robot, we implement an adaptive mode-switching algorithm comprising long-distance and close range following modes, as shown in Fig 3. Users can dynamically select modes via a mobile interface, tailoring the robot's behavior to specific needs. The long-distance mode offers remote companionship and emergency services, while the tight following mode enables continuous close-range assistance. Mode transitions are executed through the loading of mode-specific startup files, each encapsulating optimized parameters and functional configurations. This mechanism improves the AGV's adaptability and user control, facilitating efficient and reliable service across diverse scenarios in elderly care contexts.

TABLE I
MULTI-ANGLE FOLLOWING PERFORMANCE

Trajectory	Linear		L-shaped	
PresetAngle (°)	Mean (°)	Std (°)	Mean (°)	Std (°)
-40	-35.21	6.70	-34.00	13.68
-30	-25.34	10.62	-30.78	10.39
-20	-20.72	7.11	-22.99	6.98
-10	-10.99	3.65	-14.66	10.13
10	12.60	5.82	8.00	7.06
20	20.46	4.99	18.76	11.29
30	27.32	8.09	23.70	7.94
40	35.37	7.12	33.19	15.68

IV. EXPERIMENTS

In this section, the effectiveness of two close-range following methods and one long-range following method is validated and evaluated in physical environments using a differential-driven robot.

A. Experimental Results

1) Close Following Experiment: The UWB tag was placed at the target's waist center to evaluate the following distance variations. The following angle was maintained at 0 degrees, with distances set between 0.3 m and 0.7 m. To ensure safety, we imposed a maximum speed limit of 1.1 m/s. We tested both close-following algorithms under two distinct scenarios: an L-shaped trajectory and a straight-line trajectory.

We initially explored a UWB–ultrasonic configuration, which maintains distance but does not support multi-angle tracking. As shown in Fig. 4, our comparison across straight and L-shaped trajectories shows that the ultrasonic-based approach exhibits larger fluctuations—especially on L-shaped paths—due to sensor directionality and environmental interference. In contrast, the UWB+LiDAR method delivers markedly better stability and reliability, maintaining consistent tracking and small distance errors in both trajectory types. This performance gap highlights the suitability of the LiDAR-based solution for elderly-assistance scenarios that demand precise and dependable close-range following.

2) Multi-Angle Experiment: To validate angle variations during the following process, a UWB tag was positioned at the target's waist center to provide positional data. The following distance was maintained at a constant 0.5 meters, while the following angles were set to -40 to 40 degrees, respectively. Data on angle changes were collected throughout the process. Due to the suboptimal performance of the multi-angle following ultrasonic algorithm, only the multi-angle following algorithm based on LiDAR is tested.

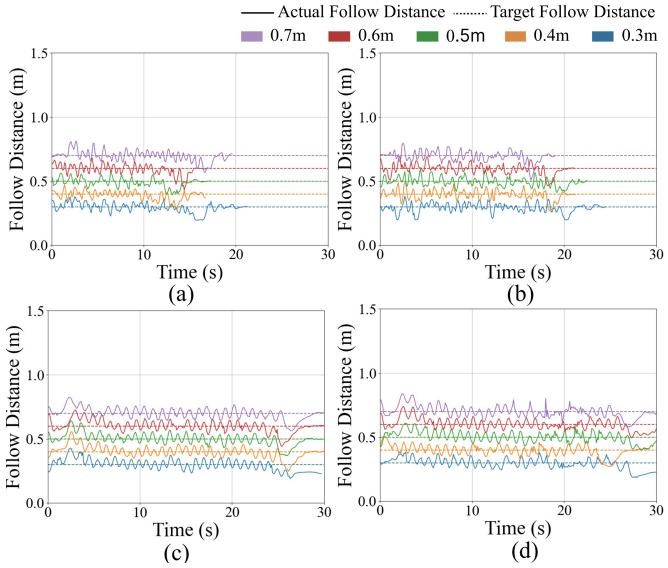


Fig. 4. Following Distance Variations: Ultrasonic+UWB (top row) v.s. LiDAR+UWB (bottom row) in Linear (left column) and L-Shaped (right column) Trajectories. This figure compares the following distance performance of two sensor fusion approaches. Panels (a)–(d) show experimental data for both linear and L-shaped trajectories. While both methods maintain a narrow error margin, the ultrasonic+UWB algorithm exhibits greater fluctuation due to sensor measurement errors. In contrast, the LiDAR+UWB approach demonstrates more consistent and robust tracking performance.

TABLE II
'PERFORMANCE COMPARISON: UWB-LIDAR CLOSE-RANGE FOLLOWING' V.S. 'FOLLOWING CLOSELY' METHOD

Method	Trajectory	FD (m)	Mean (m)	Std (m)
Following Closely [14]	Linear	0.30	N/A	N/A
		0.40	0.874	0.178
		0.50	0.936	0.126
		0.60	0.983	0.182
		0.70	0.959	0.142
	L-shaped	0.30	N/A	N/A
		0.40	0.828	0.244
		0.50	1.063	0.144
		0.60	1.038	0.201
		0.70	1.071	0.133
Close Range Following (Ours)	Linear	**0.30**	**0.297**	**0.045**
		0.40	**0.400**	**0.046**
		0.50	**0.500**	**0.046**
		0.60	**0.600**	**0.048**
		0.70	**0.700**	**0.045**
	L-shaped	**0.30**	**0.291**	**0.044**
		0.40	**0.396**	**0.043**
		0.50	**0.500**	**0.043**
		0.60	**0.600**	**0.043**
		0.70	**0.699**	**0.047**

As shown in Table I, the mean following angles closely approximate the preset values, with standard deviations generally within ±10 degrees. However, some preset angles exhibit larger variations, particularly in the L-shaped trajectory. These fluctuations can be attributed to two main factors: First, the UWB directional angle error is ±5 degrees. Second, the UWB distance error is ±10 cm. Since the robot's following direction depends on both angle and distance, these errors cause larger fluctuations in the following angle. Our algorithm supports multi-angle following modes, enhancing the robot's adaptability and versatility, whereas the "Following Closely" [14] method does not support multi-angle following.

TABLE III
LONG-DISTANCE FOLLOWING RECALL PERFORMANCE

Trigger Mechanism	Button Recall		Fall Detection Recall	
Distance (m)	Success Rate (%)	Recall Time (s)	Success Rate (%)	Recall Time (s)
10	100%	9.63	100%	9.68
15	100%	13.59	100%	13.88
20	100%	17.86	100%	17.99
25	90%	22.14	80%	22.46
30	80%	26.48	70%	26.55

3) Comparative Experiment of Close-Range Following: In addition to the original experimental setup, comparative experiments were performed by implementing the method proposed in [14] on our robot platform. This method was chosen for comparison as it most closely resembles our system's capabilities in close-range following, according to a thorough review of existing literature.

Table II compares our UWB-LiDAR Close-Range Following method with the "Following Closely" method [14]. For linear trajectories, our approach demonstrated significantly higher precision, maintaining mean distances of 0.30–0.70 m with low standard deviations 0.04 m, whereas the "Following Closely" method exhibited larger fluctuations. The difference was more pronounced in L-shaped trajectories, with the "Following Closely" method showing increased variability , while our UWB-LiDAR fusion maintained superior stability. These results indicate our method's suitability for applications requiring reliable close-range tracking within 0.50 m, particularly relevant for elderly care scenarios. Notably, "Following Closely" does not support operation at 0.30 m, which is central to our study. These results indicate the suitability of our method for reliable close-range tracking in elderly-care scenarios.

4) Long-Distance Following Experiment: We evaluate long-distance recall at 10–30 m in 5 m increments. For each distance and each mechanism—button trigger and fall detection—we conduct 20 independent trials under a fixed protocol. During recall, the robot follows a segmented speed profile (1.2 m/s for $d > 2$ m; 0.6 m/s otherwise) with randomly placed obstacles. As summarized in Table III, button-triggered recall achieves 100% success up to 20 m and 80% at 30 m; fall-triggered recall similarly reaches 100% up to 20 m and 70% at 30 m. Obstacle avoidance succeeds in 100% of trials. Across all distances, the robot reaches the target promptly for both triggers, indicating real-time responsiveness and stable mode switching.

B. Limitations and future work

Although the initial multi-sensor design may appear complex, its long-term benefits in safety, responsiveness, and adaptability outweigh the overhead in dynamic elderly-care settings. To address deployability, the PC can be replaced by an embedded platform (e.g., Jetson). The LiDARs can be swapped for lower-cost models. The smartphone interface may be replaced by a button-style Bluetooth (BLE) device for command transmission, which enhances deployability and ease of use. While our current evaluation is conducted in controlled settings, supplementary video provides a scenario demonstration with pedestrian interference, the video shows that the system maintains stable performance under dynamic-target interference. Future work will include large-scale, multi-scenario trials with diverse obstacles to quantify tracking accuracy.

V. Conclusion

We introduced a novel full-range and multi-angle following system that addresses the diverse and critical needs in elderly care, ranging from close companionship to privacy-preserving distant monitoring. This system bridges the gap in the research of the person-following robots by integrating both close-range and long-distance following capabilities, offering a more comprehensive solution than existing approaches. It strikes a balance between privacy preservation and effective monitoring, catering to the specific needs of elderly care scenarios. Our system significantly extends the capabilities of traditional person-following robots by offering an integrated solution that spans from close-range assistance to long-distance monitoring. Experimental results demonstrate the system's effectiveness across various distances and use-case scenarios. This versatility not only improves the safety and independence of older adults but also holds the potential to transform elderly care practices in diverse living environments.

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